

Compile as Hypothesis: A Headroom Typology for Prompt Optimization on Real Reasoning Models

Anonymous ACL submission

Abstract

Prompt compilers re-search or re-write prompts per model, implicitly assuming the compiled artifact is a deployable improvement. We evaluate this assumption under paired, same-day measurement on real reasoning models and find it mostly false—and we characterize *when* it fails. We present APC (Adaptive Prompt Compiler), a profile-guided prompt compiler whose outputs are treated as testable hypotheses rather than defaults: every compiled prompt runs in parallel with the uncompiled base at equal budget, and the deployment prior is a measured quantity that can be negative. Across a four-task deterministic simulator and a real-LLM phase spanning three tasks, five external benchmarks, and four independent optimizer families (APC genome search, its profile-guided variant, GEPA, ESPO), accuracy gains of structured prompt search on strong reasoners are statistically null, while a discrete structural genome robustly prevents seed-collapse. From 190-problem runs on a reachable knowledge-type benchmark we distill a *headroom-type taxonomy*: physical-truncation and knowledge-type headroom are unreachable by any prompt pathway, protocol/format headroom is reachable and favors structured genomes, and the discriminator should run before any prompt optimization is attempted. We then surface a genuine cross-model divergence: on gpt-6, free-text reflective optimization (GEPA) gains +.14 on a task where structured search has none—decomposed into a quantified judge-gaming component ($\sim.035$) and genuine task-instruction improvement ($\sim.105$)—while the gated ESPO baseline stays null on the same cell. A deployment gate (AutoAPC-Select) turns these findings into a selection rule with zero-regret rescue of seed-collapse arms. We release the full harness, all row-level traces, and a per-call backend-verification mechanism motivated by a multi-backend gateway contamination incident we document in detail.

1 Introduction

The same business task requires different prompts on different LLMs: format sensitivity is model-specific (Sclar et al., 2024), and skill profiles differ per model (Chen et al., 2023; Liang et al., 2023). Existing evaluation suites rank models but do not guide prompt construction; existing prompt compilers such as DSPy (Khatab et al., 2024) re-search prompts per model but keep the compilation prior implicit and uninterpretable. In practice, a deployed compiled prompt is an untested bet: if the compilation was harmful, nothing in the pipeline notices.

This paper makes the compilation prior *explicit, measurable, and falsifiable*. Our principle, **Compile-as-Hypothesis**, states that the output of a prompt compiler is a hypothesis to be tested, not a deployable default: prior value is a measured quantity that can be negative. Operationally, the pipeline always runs two roots in parallel at equal budget—a profile-rule root (APC-FULL) and the uncompiled base root (APC-SAFE)—and deploys the winner. The prior value is $\Delta(\text{rule-root}, \text{base-root})$, and it can be negative.

We evaluate this protocol at three levels. (1) A deterministic four-task simulator (APCBench) with three model profiles, giving exact ground truth and statistically powerful paired comparisons. (2) A real-LLM phase on qwen3.8-flash spanning three business tasks, five external author-unseen benchmarks (GSM8K (Cobbe et al., 2021), MATH Level-5 (Hendrycks et al., 2021), AIME 2024+25, GPQA-Diamond (Rein et al., 2024), HLE-exact (Phan et al., 2025)), and official reproductions of the two strongest public baselines: GEPA (Agrawal et al., 2025) and ESPO (Anonymous, 2026b). (3) A multi-provider replication matrix on gpt-6 and a second qwen provider, with per-call backend verification.

Our findings are largely *negative, and the negatives are the contribution*. On strong reasoners,

accuracy gains of structured prompt search are statistically null: the base-root arm holds the zero-shot floor on all three real tasks, the profile-rule root is a net liability (robust across three seeds, including a bimodal collapse to .13), official GEPA and ESPO reproductions land within the same-day noise band, and a token-starved discriminative experiment pins six optimization pathways at the judge floor simultaneously. External author-unseen benchmarks reproduce both findings, with the model saturating every benchmark (AIME: 60/60). What survives is structural: a discrete genome prevents seed-collapse, cross-task zero-adaptation transfer reaches zero-shot quality at zero budget (up to $+0.79$ over cold re-search), and a pre-registered deployment gate (AutoAPC-Select) rescues both seed-collapse arms at zero regret.

The decisive finding is a **headroom-type taxonomy**. In a 190-problem HLE-exact subset with genuinely reachable knowledge-type headroom, all six optimization arms tie; on GPQA-Diamond ($n=190$) the tie replicates and reflective search accepts zero candidates in 28 iterations—even though reflection demonstrably learned domain content rules. Prompt-optimization payoff is discriminated not by whether headroom exists but by *what kind*: physical-truncation and knowledge-type headroom are out of reach for any prompt pathway (the ability lives in weights), while protocol/format headroom is reachable and favors structured genomes. Deployment should begin with a headroom-type diagnostic, and in knowledge-type scenarios the correct action is to change the model, not the prompt.

We then report a genuine **cross-model divergence** that sharpens the taxonomy: on gpt-6, free-text reflective optimization (GEPA) gains $+0.14$ on financial where structured search has none. Forensic decomposition of the evolved prompt shows the reflective optimizer reverse-engineered our rule judge from per-dimension scores (a judge-gaming component worth only ~ 0.035 of the gain), while the remainder comes from genuine task-instruction improvement. The gated ESPO baseline, run under full backend enforcement on the same cell, stays null—so the divergence is specific to unconditional best-of-search. Finally, a multi-backend gateway routed some evaluation calls to a different model during failure windows; we detect this from response metadata, ship a per-call backend-verification mechanism, and withdraw the affected cells rather than claim them.

Contributions.

- **C1** Two-root compilation protocol that makes the compilation prior measurable and negative-capable (§3), with 41 unit tests and CI-ordered analyses.
- **C2** A headroom-type taxonomy distilled from 190-problem runs, with a pre-deployment diagnostic (§5.5).
- **C3** Baseline null coverage closed by official GEPA/ESPO reproductions plus a pre-registered token-starved discriminative experiment (§5.4).
- **C4** A cross-model divergence discovery with judge-gaming decomposition, and an evaluation-infrastructure forensics protocol (per-call backend audit) (§6).
- **C5** AutoAPC-Select, a deployment gate with 5/7 exact-oracle and zero-regret rescue of seed-collapse arms (§7).

2 Related Work

Prompt optimization. APE (Zhou et al., 2023), OPRO (Yang et al., 2024), and ProTeGi/textual gradients (Prasad et al., 2023) search free-text instruction spaces; DSPy (Khatab et al., 2024) compiles declarative pipelines per model; GEPA (Agrawal et al., 2025) performs reflective prompt evolution from failure minibatches; ESPO (Anonymous, 2026b) selects among strategy-typed rewrites via bootstrap. These systems implicitly treat the optimized prompt as a deployable. We invert this: the artifact is a hypothesis tested against the uncompiled base under paired measurement, and the interesting quantity is the *prior* of compilation. Anonymous (2026a) report capability-dependent compilation gains consistent with our profile-liability finding.

Prompt sensitivity and evaluation reliability. Sclar et al. (2024) quantify format sensitivity; Wu et al. (2024) show evaluators reward style over substance. Both motivate our hardened exact-match judge and the paired same-day design: with temp= 0 reasoning models exhibiting cross-day score drift, all comparisons in this paper are paired same-day within a quantified noise band.

Test-time compute. Snell et al. (2024) study optimal test-time budget allocation for reasoning. Our budget accounting is the prompt-side analogue: optimizer meta-calls are not charged, and the budget

currency is task rollouts, mirroring GEPA’s protocol for comparability.

Benchmarks. We use GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), MMLU-style MCQ constraints, GPQA-Diamond (Rein et al., 2024), HLE-exact (Phan et al., 2025), and rank-oriented suites (HELM (Liang et al., 2023), FLASK (Chen et al., 2023)) only as context: they rank models but do not decide whether to deploy a compiled prompt.

3 Method

3.1 Structural genome and task specification

A PROMPTGENOME is a discrete, re-compilable prompt representation: role, constraints, reasoning directives, few-shot example blocks, output directives (schema strictness, inclusion of schema in prompt), and verification hooks. A TASKSPEC declares the task contract (input mapping, gold schema, per-dimension scoring weights). Unlike continuous soft prompts, a discrete genome is model-agnostic: it renders to any renderer and can be transferred across models zero-shot (§5.3).

3.2 Two-root compilation

The compiler (APC-FULL) applies profile-derived rules to the base genome: loci are set from capability-gated priors (e.g., few-shot blocks added only when the profile’s few-shot benefit is low; schema directives strengthened when JSON reliability is weak). The base root (APC-SAFE) is the uncompiled genome. Both run in parallel at equal budget on dev, the winner deploys. This makes the compilation prior = $\Delta(\text{rule-root}, \text{base-root})$ a measured, signed quantity.

3.3 Search

Evolution runs two generations with population 8, elite 3, and a seeded RNG; mutation selects a locus from the genome’s search space and an alternative value uniformly (GENOMEMUTATOR), or roulette-sampled with profile-prior weights plus elite-reinforced bandit correction (PROFILEGUIDEDMUTATOR, PGAM). Successive halving is available for large search spaces. The budget currency is task rollouts (dev examples scored); optimizer meta-calls are not charged, matching GEPA’s accounting.

3.4 Judge

Ground truth is a hardened exact-match judge (v3.2): answer normalization, multiset comparison for notation-variant equivalence, adversarial format checks; 27 positive and 4 negative regression cases with full prediction traces persisted for recomputation. Human auditing of 12 failure samples drove three judge revisions. Judge reliability is quantified in §7.

3.5 Simulator

APCBench is a deterministic four-task simulator (financial analysis, contract extraction, math reasoning, constraint following) $\times$ three model profiles, giving exact ground truth for powerful paired comparisons without API noise. All simulation seeds are fixed; analyses are CI-ordered (adding analyses does not change existing confidence intervals).

4 Experimental Setup

Real APIs. qwen3.8-flash via DashScope (the primary model; whitelist credential) and gpt-6 via an OpenAI-compatible gateway. All runs are temp= 0 with model-side reasoning enabled.

Protocol. dev 5 / val 8 / holdout 20 examples, search budget 8, seed 42 unless stated. Every comparison is *paired same-day*: a quantified same-day noise band of $\pm .007$ and a cross-day drift of .028 were measured from repeated z0 evaluations; any “gain” within $\pm .01$ is judged noise.

Arms. zero-shot (base genome, no compilation), MANUAL (hand-written prompt), APC-FULL (rule-root search), APC-SAFE (base-root search). Base-lines: GEPA (official Algorithm 1 reproduction: failure-driven reflective rewriting, per-instance Pareto parent sampling, full-val champion selection) and ESPO (Diagnose/Propose/bootstrap-Select).

External benchmarks. GSM8K, MATH Level-5, AIME 2024+25 (60 problems), GPQA-Diamond (190 holdout problems), and a stratified 90-problem HLE-exact subset selected for reachable knowledge-type headroom (zero-shot accuracy $\approx .10$).

Contamination control. For the multi-provider phase, every API call’s response model field is audited to a per-call stream, and a configured expectation (expect_backend) raises a retryable

Task	zero-shot	manual	apc-full	apc-safe (b6, s42)	cold	t0	warm-start
financial	.6666±.0023	.6685±.0018	.3085±.2362*	.6664±.0047	.6402	.8439	.8441
contract	.9771±.0001	.9775±.0002	.7035±.4586*	.9771±.0003	.1850	.9778	.9582
math	.8436±.0001	.8438±.0001	.7769±.1164	.8436±.0001	.6336	.8443	— [†]

Table 1: Real qwen3.8-flash, multiseed mean±SD (holdout; $n=3-6$ seeds per cell, seeds 43–47 + s45 statics). *bimodal collapse: rule-root search falls toward the judge floor in 3/5 (financial) and 1/3 (contract) seeds; base-root never collapses in 12/12 runs. gpt-6 clean-cell analogues: zero-shot .6862, MANUAL .6971 (search arms withdrawn, §6.3).

Table 2: Cross-task transfer holdout scores. [†]withdrawn: served by a fallback backend during a gateway failure window (§6.3).

of a cross-task champion genome: 0 budget, $\geq$ same-task zero-shot, up to +.79 over cold re-search.

error on mismatch; cells that cannot be verified are withdrawn from claims (§6.3).

5 Results

5.1 Simulator: structured search helps exactly where it should

The simulator phase yields three clean readings. Genome search significantly beats zero-shot on all four tasks (+.009 to +.029, CIs excluding zero); evolution beats random search on compositional tasks (+.003) but loses on single-locus tasks (−.013, successive-halving selection noise); PGAM is indistinguishable from uniform mutation (pooled +.0004, a null result); migration reaches native quality at 30% of the re-search budget.

5.2 Real qwen3.8-flash: the base-root holds the floor

On all three real tasks (Table 1), the base-root arm holds the zero-shot floor within −.004, while the profile-rule root is a net liability: dev-score deficits of 0.2–0.53 that are only partially recovered within budget. A five-seed replication sharpens this into the paper’s strongest structural claim. Across seeds 43–47, APC-FULL exhibits *bimodal catastrophic collapse* on every task: financial .5704/.5705/.1334/.1318/.5639/.1428 (mean .3085, SD .2362; 3/5 runs collapse toward the judge floor), contract .9782/.9583/.1741 (mean .7035, SD .4586), and math .8442/.8440/.6425 (one −.20 drop). Meanwhile APC-SAFE *never* collapses: 12/12 runs across the three tasks sit within SD $\leq$.005 of the zero-shot floor (financial .6664±.0047, contract .9771±.0003, math .8436±.0001). The two-root protocol is therefore not methodological hygiene but insurance against a $\sim$ 50% catastrophic failure mode of rule-root search. Under the headline protocol, the strongest deployment policy is *zero-adaptation transfer*

5.3 Cross-task transfer

The strongest deployment policy is zero-adaptation transfer of a cross-task champion genome: 0 adaptation budget, at or above target zero-shot, up to +.79 over cold re-search (qwen). On gpt-6 the new pair replicates with clean cells only (Table 2); the warm-start arm is withdrawn ([†], §6.3).

5.4 Baselines and the deployment gate

The official GEPA reproduction lands within the paired same-day noise band on financial (+.0024), and ESPO likewise (.7009)—four independent optimizer families (APC genome, PGAM, GEPA, ESPO) all null in the same regime, so the null is not an artifact of APC’s representation. AutoAPC-Select (F8) deploys `val-argmax` with an Occam tie-break inside the noise band: on 7 replay+blind groups it is 5/7 exact-oracle, and both seed-collapse arms are rescued at zero regret (the s45 rescue is a forward blind test under the frozen gate). The gate’s only failure mode (small-val overestimation, +.0196) is characterized.

5.5 Headroom-type discrimination

Three real experiments discriminate *which* headroom is reachable:

F10: token-starved (pre-registered). With `max_tokens` 150 the model collapses to .15; all six optimization pathways—structured, reflective, free-text—score at the judge floor on the same day. When output capacity is physically truncated, no prompt pathway can win: the ability lives in weights, not the prompt.

F11: reachable knowledge-type headroom. On a 90-problem HLE-exact subset with genuine headroom (zero-shot accuracy .10, ceiling .3+), all six arms tie (.388–.412, 2–3/30). A confirmatory

Provider	zero-shot	manual	full	safe
DashScope (s45)	.6692	.6665	.1318	.6657
alias provider (s42)	.6669	.6665	.6150	.6694
gpt-6 (s42)	.6862	.6971	— [†]	— [†]

Table 3: Financial holdout across providers. [†]withdrawn: served by an anonymous fallback backend during a gateway failure window; re-runs abandoned per decision (§6.3).

$n=100$ round replicates the tie (GEPA vs zero-shot McNemar $p=.219$) and measures the same-text nondeterminism floor at 2.7%. GEPA’s reflection demonstrably learned domain content rules (SMILES conventions, perturbation-theory regime qualifiers) that *transferred zero* to the holdout.

GPQA-Diamond ($n=190$). The tie replicates on a second real dataset: five arms .853–.868, reflective search accepts zero candidates in 28 iterations ($p=.219$ class evidence); the champ holdout re-evaluation lands at .8856, within same-text repeatability (+.0127) of zero-shot .8729. Synthesis: with knowledge-type headroom, prompt optimization is the wrong lever; the correct deployment action is a model change.

5.6 Multi-provider replication

Table 3 replicates the financial matrix across two independent qwen providers (DashScope; and a gateway endpoint that probe-verified as a qwen3.8-flash alias) plus clean gpt-6 cells. Across both qwen providers no search arm beats the static floor beyond the noise band, confirming the regime classification is provider-independent for this model. The seed-42 financial champion genome is byte-identical (md5-verified) across both providers: the seeded mutation trajectory is provider-independent and the fitness ranking coincided at every selection decision.

6 Analysis

6.1 A cross-model divergence: free-text search finds what structured search cannot

The qwen-era GEPA round measured a same-day null on financial (+.0024). On gpt-6, the identical protocol yields hold **.8914** vs the paired same-path z0 **.7039**—a +.14 gain on the one task where structured search showed zero headroom. Forensic decomposition of the evolved prompt: its style-rules section is an explicit encoding of the judge’s rubric (a two-template sum-

mary constraint copied from the scoring dimensions, metrics-inclusion rules, and “violations score zero” wording)—the reflective optimizer reverse-engineered the judge from per-dimension failure traces. Stripping that section costs only $-.035$ (.8441 verbatim $\rightarrow$.8092 stripped); the remaining $\sim+.105$ comes from task-instruction rewriting that lifts judge-accuracy .41 $\rightarrow$.68–.75. Three readings follow. (i) The no-headroom conclusion is *space-specific*: structured mutation has no headroom, free-text reflective rewriting does, and the gap is model-dependent (null on qwen, +.14 on gpt-6). (ii) Free-text optimizers without structural safeguards do game rule judges, but gaming was a minority of the measured gain. (iii) The two-root paired protocol is exactly what exposed this: an unconditional deploy would have shipped a partially gamed prompt. The gated ESPO baseline, re-run under full backend enforcement on the same cell, declines to adopt any candidate and stays null (.7021 vs .7039)—the divergence is specific to unconditional best-of-search.

6.2 PGAM’s prior fails to transfer

The simulator’s pooled PGAM-vs-uniform null (+.0004) was tested on gpt-6 in the one cell with search variance (math). The profile-guided mutator cold-searches to **.5960** vs uniform .6336 ($-.0376$, outside the band). The gpt-6 profile marks few-shot benefit 0.0, so the prior up-weights example genes $3\times$ —and few-shot examples hurt a frontier reasoner that needs none; two generations of bandit correction cannot undo a wrong prior inside budget 8. Profile-weak dimensions are not mutation-productive dimensions on strong models: the defect-compensation prior is a simulator-era hypothesis that fails its first real-model test in the negative direction.

6.3 Evaluation infrastructure: two contamination forensics

Gateway backend mixing. The gpt-6 gateway is a multi-backend aggregator: during failure windows it re-routes the endpoint, and response model fields prove five evaluation cells in our matrix were served by an anonymous fallback backend while others were clean gpt-6. A dedicated endpoint that we had treated as a second frontier model probe-verified as a qwen3.8-flash alias (3/3 response models; score parity with the DashScope history, e.g. manual .6665 identical). Remediation: a per-call response-model audit stream

and `expect_backend` enforcement (mismatch raises a retryable error). The enforced re-run of the contaminated cells aborted correctly inside a fallback window (three mismatch aborts, zero contaminated rows written); the clean gpt-6 window then remained unavailable for over 24 hours, and per decision the re-runs were abandoned—the affected cells are withdrawn ($\dagger$), not claimed. The clean-cell claims that stand: contract saturation and search liability, zero-adaptation transfer (+.2107), GEPA divergence direction, and the PGAM negative result.

Service drift. A dedicated same-text floor run on GPQA (seed 927) scored .6501 vs .8729/.8856 for the same 931-char text measured two days earlier. Same-problem agreement was 57.7% with one-sided flips 74:6, uniform across the run; a live probe reproduced the earlier correct behavior 4/4. We therefore attribute the drop to a provider-side serving change, not to prompt nondeterminism, quarantine the run, and carry the floor as the +.0127 same-text repeatability from the two healthy-window runs. This is the same class of risk [Zheng et al. \(2023\)](#) document for judge stability, here on the serving side.

7 A Deployment Gate

AutoAPC-Select (F8) operationalizes the findings: deploy `val-argmax`, break ties inside the noise band toward Occam (fewer moving parts), and never deploy a compiled artifact that failed its paired test. On 7 replay+blind groups the gate is 5/7 exact-oracle, both seed-collapse arms are rescued at zero regret (including one forward blind test under the frozen gate), and the single failure mode (small-val overestimation, +.0196) is characterized. Combined with the headroom-type diagnostic—run *before* any optimization: if the regime is knowledge-type or truncation-type, change the model instead—this converts the taxonomy into a deployment procedure: (1) diagnose headroom type; (2) if protocol/format-type, run two-root compilation with paired measurement; (3) deploy only gate-passing artifacts; (4) audit serving backends per call.

8 Limitations

Ground truth. Rule-judge ground truth is authored; an independent third-party judge and a 50–100-case human spot-check are missing (the judge’s 27+4 regression suite and full prediction

traces mitigate but do not eliminate this). Open-instruction tasks without unique answers (judge-dependent) are out of scope.

Seeds and cells. All qwen real-LLM cells are multiseed ($n=3-6$; Table 1 reports $\text{mean} \pm \text{SD}$); new-model (gpt-6) clean cells remain single-seed with no CI (the $\pm .007$ band is the qwen-era prior). The gpt-6 search cells were withdrawn after gateway backend mixing (§6.3) and are not replaced: the genome-search null on gpt-6 is therefore *untested*, not established—only the qwen null (two providers, multiseed) is established. The GEPA divergence is single-seed; its direction is robust but its exact magnitude needs more seeds.

Baselines and models. The SEPO head-to-head is deferred (no real-API run); the structured-editing family is covered indirectly by mechanism argument. A second frontier provider with per-call-verified identity is required to complete the multi-model search matrix. PGAM’s negative result is one cell; genuinely multimodal real tasks with heterogeneous profiles remain untested.

Internal validity of simulation. APCBench is self-built; its role is controlled comparison, not external validity, which is carried by the real-API phase and external benchmarks.

9 Ethical Considerations

All ground truth in the judge path is authored; we flag this as the paper’s principal validity risk rather than burying it, and release the full prediction traces so any reader can re-judge. The evaluation-infrastructure forensics (§6.3) documents third-party serving instability without identifying vendors beyond what response metadata discloses. No human subjects, no personal data, and no model training are involved.

10 Conclusion

Treating compiled prompts as hypotheses rather than defaults changes what an evaluation finds. Under paired same-day measurement, structured prompt search on strong reasoners is statistically null across four optimizer families, and the profile-rule prior is a net liability that only a discrete base-root parallel run prevents from collapsing. The payoff question is decided by headroom type, not headroom existence: a pre-deployment diagnostic, a gate that never ships a failed hypothesis, and per-call backend verification are the components

we release. Free-text reflective search on gpt-6 breaks the null—and its forensic decomposition shows exactly why unconditional adoption is dangerous: part of the gain was our own judge, reverse-engineered. We believe this evaluation protocol, with its negatives intact, is the deployable contribution.

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A Reproducibility Summary

One-command reproduction: 601
 bench_apc.py [25|50|100] plus 602
 bench_contract/math/follow/transfer.py 603
 and eval_robustness.py (fixed 604
 seeds, deterministic, no network). Real- 605
 API harness: bench_real_full.py 606
 / bench_real_transfer.py 607
 / bench_real_gepa.py 608
 / bench_real_espo.py 609
 / bench_real_hle.py / 610
 bench_real_pgam.py, with row- 611
 level streams, per-call backend audit 612
 (artifacts/backend_audit/), and judge 613
 regression suite persisted. 41 unit tests green; 614
 results audited by a three-way (JSON/zh/en) 615
 consistency script over every number cited. 616